



3-MONTH INTENSIVE PROGRAMME

AI & Machine Learning **Bootcamp**

Go from your first line of Python to deployed machine-learning models — a hands-on, mentor-led path into AI.

FORMAT

Onsite & Online

DURATION

3 Months

FEE

NGN 250,000

About Data-Lead Africa

“We turn data into insight, and insight into decisions.”

RESEARCH · ANALYTICS · TRAINING

Data-Lead Africa is a research and training consulting firm providing monitoring & evaluation, data analytics, data science training and strategic consulting services. We combine statistics, research methods and technology to process data into insight.

Our core expertise spans research and data collection, data analysis and data science training, monitoring & evaluation, and business, human-resource and market-research analytics, delivered by a team of seasoned industry experts.

PARTNERS & CLIENTS WE HAVE WORKED WITH



Built to launch your AI career

The AI & Machine Learning Bootcamp is a comprehensive 3-month programme that gives you the foundational and practical skills to launch a career in AI.

Whether you're new to the field or expanding your analytical toolkit, you'll gain hands-on experience with the tools and techniques used by data professionals worldwide. It follows a clear path:

01

Foundations

Master Python, data handling and the maths behind machine learning.

02

Machine Learning

Build, evaluate and tune real predictive models.

03

Modern AI

Train neural networks and work with deep learning and GenAI tools.

04

Deploy & Launch

Ship a capstone model and step into an AI or data role.

What you'll be able to do

- Write clean, functional Python for data and AI tasks
- Build, evaluate and tune machine-learning models
- Deploy a model and present an end-to-end capstone project
- Clean, analyse and query real datasets with pandas and SQL
- Train neural networks and use modern AI and LLM tools

— HOW IT WORKS

A programme that builds real experience

Five ways we take you from learning to doing.



Recorded video lessons

High-quality, easy-to-follow video content you can learn from at your own pace.



Onsite & online

Join and collaborate with other learners both in person and online.



Peer-review sessions

Share knowledge with fellow learners and gain new perspectives on problem-solving.



Real-world case studies

Apply what you learn to solve related business problems on live projects.



Capstone project

Build and present an end-to-end AI/ML project to complete the programme.

DURATION

3 Months

DELIVERY

**In-person &
Online**

LOCATION

Abuja, Nigeria

— THE TOOLKIT

Tools you'll master

You'll graduate fluent in the tools AI and machine-learning practitioners actually use on the job.

Python

Jupyter

NumPy

pandas

scikit-learn

TensorFlow

Keras

SQL

Power BI

Matplotlib

Streamlit

Git

— CURRICULUM

Four core areas, one capstone

Programming & Data

Python from beginner to advanced, with pandas and NumPy for real data.

Databases & SQL

Store, structure and query relational data with SQL.

Machine Learning & AI

Core ML, deep learning, NLP and generative AI.

Visualise & Deploy

Build dashboards and ship trained models into production.

Modules 01–04

01

Python Programming Beginner → Advanced

Start from zero and build up to writing clean, reusable Python for real data and AI tasks.

- Setup with Jupyter & Google Colab; variables, data types, operators
- Core data structures: lists, dictionaries, sets and tuples
- Error handling, file input/output and working with APIs
- Control flow, loops, functions and scope
- Object-oriented programming, modules and packages
- Advanced Python: comprehensions, generators and decorators

02

Data Analysis pandas & NumPy

Manipulate, clean and explore real datasets with Python's core data libraries.

- NumPy arrays and fast vectorised computation
- Loading data from CSV, Excel, JSON and SQL with pandas
- Cleaning: missing values, duplicates, data types and outliers
- Filtering, grouping, merging and reshaping data
- Exploratory data analysis (EDA) and summary statistics
- Feature engineering foundations for modelling

03

SQL for Data Management Databases

Query and manage relational data — where most real-world data actually lives.

- Relational database and SQL fundamentals
- SELECT, WHERE, ORDER BY and aggregate functions
- Joins, subqueries and multi-table queries
- Inserting, updating and deleting records
- Connecting SQL databases to Python
- Indexing and query performance basics

04

Maths & Statistics for ML Foundations

The intuition behind the algorithms — just enough maths to build models with confidence.

- Descriptive statistics and probability
- Distributions, sampling and hypothesis testing
- Correlation and inferential statistics
- Linear algebra essentials: vectors and matrices
- Gradient and optimisation intuition
- Bias, variance and the maths of model fit

Modules 05–08

05

Machine Learning Core · scikit-learn

Build, evaluate and tune predictive models with the industry-standard scikit-learn library.

- The ML workflow; train, validation and test splits
- Decision trees, random forests, gradient boosting, SVM and KNN
- Feature scaling, encoding and engineering
- Supervised learning: linear and logistic regression
- Unsupervised learning: k-means clustering and PCA
- Model evaluation, cross-validation and hyperparameter tuning

06

Deep Learning & Modern AI Neural Nets · NLP · GenAI

Go beyond classical ML into neural networks and the technology defining today's AI.

- Neural network fundamentals with TensorFlow & Keras
- Convolutional neural networks (CNNs) for computer vision
- Transformers, large language models (LLMs) and prompt engineering
- Training: activations, loss functions, optimisers and regularisation
- Recurrent networks (RNNs/LSTMs) and NLP basics
- Intro to Generative AI apps, RAG and responsible AI

07

Data Visualisation Power BI · Matplotlib

Turn data and model results into dashboards that decision-makers act on.

- Visualisation best practice and data storytelling
- Power BI: connect, clean and model data with Power Query
- Publishing and sharing via Power BI Service
- Charting with Matplotlib and Seaborn in Python
- Interactive dashboards, DAX measures, slicers and filters

08

Deployment & MLOps Ship Your Models

Move models out of the notebook and into the real world, where they create value.

- Saving, loading and versioning trained models
- Containerising models with Docker (introduction)
- Collaborating with Git and GitHub
- Building APIs and web apps with Flask and Streamlit
- Model monitoring and the ML lifecycle (MLOps basics)



Capstone Project

Bring it all together in an end-to-end machine-learning project on a problem of your choice — from raw data to a deployed model — presented to complete the programme.

— REQUIREMENTS & ENROLMENT

Ready to join?

Laptop specifications

Operating system **Windows, macOS or Linux**

Memory (RAM) **8 GB min · 16 GB recommended**

Storage **256 GB SSD**

Processor **Modern quad-core or better**

No high-end GPU required — heavier model training runs on free cloud notebooks such as Google Colab.

Who should apply

Beginners and professionals moving into AI, machine learning or data science — graduates, analysts, developers and researchers who want practical, job-ready skills.

How to enrol

- 1 Register**
Complete the short enrolment form to express your interest.
- 2 Secure your seat**
Make the one-time payment to confirm your place in the cohort.
- 3 Onboard & begin**
Get your welcome pack, join the community and start learning.

ONE-TIME FEE

NGN 250,000

Covers the full 3-month programme: video lessons, live sessions, peer reviews, case studies and your capstone.



“Ready to accelerate your career?”

Registration is open. Secure your seat in the next AI & Machine Learning cohort and start building real, job-ready skills.

ENROL NOW

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ONE-TIME FEE

NGN 250,000

Data-Lead Africa

Plot 759, Bassan Plaza, Central Business District, Abuja
info@dataleadafrica.com · +234 703 050 0741 · dataleadafrica.com

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@DATA-LEAD AFRICA

A decorative graphic at the bottom of the page consisting of several concentric, semi-transparent orange circles of varying sizes, creating a ripple effect.